

# Georges Parfait Djimefo Kape | AI Developer | Agentic AI & Cloud Systems

Montréal, Québec, Canada  
📞 +1 514-245-3182 • ✉️ gpdk2025@gmail.com

## Profile

AI developer with hands-on experience designing and deploying **production-grade agentic AI systems**, **multi-agent LLM workflows**, and **enterprise-scale RAG chatbots** on AWS. Strong background in **AI evaluation & benchmarking**, **cloud-native architecture** (serverless, containers), **CI/CD automation**, and **ETL/ELT data pipelines**. PhD expected Summer 2026 with research in **AI**, **graph neural networks**, **multi-agent reinforcement learning**, and **cybersecurity**. Proven ability to communicate technical trade-offs to executives and non-technical stakeholders. Innovation, teamwork, leadership, and results-oriented.

## Technical Skills

**LLM / Agents:** LangChain, LangGraph, LangSmith, MCP, RAG (FAISS), CRAG, RAGAS, Claude (Haiku, Opus), Kiro  
**AI / ML:** PyTorch, TensorFlow, Scikit-Learn, MARL (QMIX, QTRAN), GNN, DQN, NLP, Computer Vision, LLMs  
**Cloud:** AWS (SageMaker, Bedrock, EKS, Lambda, CloudWatch, S3, ECR, CloudFormation), OpenStack, Kubernetes, Docker  
**DevOps / IaC:** CI/CD (GitHub Actions, CodeBuild, OIDC), Git, Jenkins, GitLab, Terraform, Infrastructure as Code  
**Data:** PySpark, Databricks, Snowflake, Delta Lake, SQL, Relational Databases, ETL/ELT  
**Languages:** Python, Java, C++, PHP, SQL, R, JavaScript, TypeScript  
**Web / APIs:** REST APIs, FastAPI, Streamlit, React, Angular, Node.js, D3.js, Plotly, Dash, Power BI, Tableau  
**Methods:** Agile/Scrum, TDD, Design Patterns, Statistical Modeling, MLOps, AI Governance

## Relevant Experience

### Ericsson

Montréal

*ML Intern — BCSS Group, Agentic AI & Cloud Systems*

2026

- Designed and deployed a **production RAG Autonomous Agentic AI system** that automates telecom radio unit log anomaly detection, replacing manual review for enterprise clients.
- Built a **multi-agent LLM workflow**: 6-tool ReAct agent orchestrated by Claude Haiku 4.5 on Bedrock (SageMaker scoring, FAISS RAG, RF physics world model, pattern classification, CloudWatch health monitoring), replicating analyst reasoning.
- Evaluated and benchmarked foundation models with **RAGAS** faithfulness/relevancy scoring and evidence-based **meta-cognition** confidence grading (HIGH/MEDIUM/LOW) to optimize performance-cost trade-offs.
- Trained a custom **Dual-Attention SLM** achieving **F1 90.6%**, 99.97% accuracy, and **1 ms/log** inference latency on AWS SageMaker.
- Engineered a **production-scale map-reduce pipeline**: triage → cluster by failure signature → parallel agent investigation → adaptive cross-cluster synthesis. Handles 120+ anomalies in ~12 min.
- Built an **executive-facing Streamlit dashboard** exposing accuracy, usage, latency, and cost-per-query metrics for leadership decisions.
- Deployed as a **K8s pod on EKS** via Cloudflare tunnel; built an **event-driven pipeline** (S3 → Lambda → SQS → K8s worker → S3 → SNS alerts) for high availability.
- Designed a **fully autonomous retraining loop**: drift-triggered + weekly via EventBridge, with auto-promotion only when new model F1 exceeds production baseline (AI governance & monitoring).
- Implemented a **7-stage CI/CD pipeline** (GitHub Actions + CodeBuild with OIDC): test → OWASP security scan → CloudFormation infra → staging gate → deploy → smoke test → auto-rollback. Zero static IAM keys.
- Authored **131 automated tests** with an AST-based adaptive test guardian ensuring 100% production function coverage.
- Stack:** Python, AWS (SageMaker, Bedrock, EKS, Lambda, SQS, SNS, S3, ECR, CloudFormation, CloudWatch), LangChain, LangGraph, LangSmith, FAISS, MCP, Claude, Kiro, Docker, Kubernetes, CI/CD, Terraform.

### Intact Financial Corporation

Montréal

*AI Developer I — Enterprise Data Systems*

2022–2024

- Ingest and process data.
- Monitor pipelines and develop automated testing solutions ensuring system reliability.
- Stack:** PySpark, SQL, Databricks, Snowflake, AWS, Shell, Git.

## Additional Experience

### Polytechnique Montréal

Montréal

*Lecturer*

2025–2026

- Teach the Python procedural programming course; communicate complex technical concepts to non-technical audiences.

### Polytechnique Montréal

Montréal

*Lecturer*

2022–2023

- Teach the Python procedural programming course.

|                                                                                                                         |                              |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Université de Montréal</b><br><i>Scientific Assistant — MOOC AI</i>                                                  | <b>Montréal</b><br>2021–2022 |
| <ul style="list-style-type: none"> <li>Design courses and practical work in supervised learning (Python).</li> </ul>    |                              |
| <b>Ministry of Tourism and Recreation</b><br><i>Contractual Statistical Economist Engineer</i>                          | <b>Cameroon</b><br>2012–2014 |
| <ul style="list-style-type: none"> <li>Analyze data</li> </ul>                                                          |                              |
| <b>Protestant University of Central Africa</b><br><i>Associate Professor of Statistics &amp; Probability</i>            | <b>Cameroon</b><br>2010–2012 |
| <ul style="list-style-type: none"> <li>Teach statistics and probability courses.</li> </ul>                             |                              |
| <b>ISSEA (Sub-regional Institute of Statistics and Applied Economics)</b><br><i>Part-time Lecturer in Data Analysis</i> | <b>Cameroon</b><br>2009–2010 |
| <ul style="list-style-type: none"> <li>Teach the Data Analysis course (R).</li> </ul>                                   |                              |

## Publications

- [1] *Energy-Efficient and Near Real-Time Routing Algorithm for Disaster Monitoring in Wireless Sensor Networks*  
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Designed **GAPF**, a novel **meta-learning** framework fusing MARL policies (QMIX & QTRAN) via a **GCN encoder** with only **1 473 parameters**.
  - Achieved  $\geq 98\%$  packet delivery,  $2\text{--}3.5\times$  less energy,  $13\times$  less memory than baselines.
- [2] *Carbon-Aware Autonomous Data Reduction and Self-Healing Routing for 6G-Integrated Disaster Sensor Networks*  
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Proposed **CALASH**, the **first protocol** jointly optimizing Lifecycle Carbon Intensity via a hybrid **Lyapunov–DQN** engine.
  - Achieved 60% longer network lifetime and 33% lower LCI than baselines.
- [3] *Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Networks*  
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Designed **CASTER-ZT**, combining **GNN-based** decision making with a formally bounded **zero-trust** admissibility shield; enterprise security and compliance by design.
  - Achieved 74–98% rogue detection with **zero false positives**, 3 ms latency within O-RAN 10 ms budget.

## Selected Projects

|                                                                                                                                                                                   |                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Intact Financial Corporation</b><br><i>Data Engineering Intern</i>                                                                                                             | <b>Montréal</b><br>2022      |
| <ul style="list-style-type: none"> <li>Manipulate data (PySpark, SQL, Databricks, Snowflake, AWS).</li> </ul>                                                                     |                              |
| <b>Polytechnique Montréal</b><br><i>Integrator Project — IATA</i>                                                                                                                 | <b>Montréal</b><br>2021      |
| <ul style="list-style-type: none"> <li>“Flight Data eXchange — Develop an initial and contextual flight data assessment tool.” (Dash, Python, React, Electron, Axios).</li> </ul> |                              |
| <b>Polytechnique Montréal</b><br><i>Undergraduate Research Initiation Intern</i>                                                                                                  | <b>Montréal</b><br>2020–2021 |
| <ul style="list-style-type: none"> <li>Voice Recognition modeling for secure bank payments; Deep Learning (Python) deployed on Android (Java).</li> </ul>                         |                              |
| <b>Polytechnique Montréal</b><br><i>NLP Project</i>                                                                                                                               | <b>Montréal</b><br>2021      |
| <ul style="list-style-type: none"> <li>Extraction of keywords from texts as well as their type (Python).</li> </ul>                                                               |                              |
| <b>Polytechnique Montréal</b><br><i>Data Visualization Project — Chaire de recherche en fiscalité</i>                                                                             | <b>Montréal</b><br>2021      |
| <ul style="list-style-type: none"> <li>Web application to compare taxes and rates by year (Plotly.js, D3).</li> </ul>                                                             |                              |
| <b>Polytechnique Montréal</b><br><i>Entrepreneur / Research Initiation Intern</i>                                                                                                 | <b>Montréal</b><br>2020      |
| <ul style="list-style-type: none"> <li>Face Recognition modeling integrating racial and gender bias; Deep Learning (Python).</li> </ul>                                           |                              |
| <b>FLEX GROUP</b><br><i>Programming Intern</i>                                                                                                                                    | <b>Laval</b><br>2019–2020    |
| <ul style="list-style-type: none"> <li>Modeling of Automatic Optical Character Recognition (Python, PHP).</li> </ul>                                                              |                              |
| <b>International Bank of Cameroon for Savings and Credit</b><br><i>Intern for Master's in Applied Statistics</i>                                                                  | <b>Cameroon</b><br>2006      |
| <ul style="list-style-type: none"> <li>Mathematical models for forecasting a bank's income and expenses (R).</li> </ul>                                                           |                              |

## Education

|                                                                     |                                        |
|---------------------------------------------------------------------|----------------------------------------|
| <b>Polytechnique Montréal</b><br><i>PhD in Computer Engineering</i> | <b>Montréal, Canada</b><br>Summer 2026 |
|---------------------------------------------------------------------|----------------------------------------|

- AI-driven routing, carbon-aware networking, and zero-trust security for 6G sensor networks.

### **Polytechnique Montréal**

**Montréal, Canada**

*Engineering Diploma in Software Engineering*

- Concentration: Artificial Intelligence — Data Sciences.

### **Advanced National School of Engineering (ENSPY)**

**Yaoundé, Cameroon**

*Master's Degree in Applied Statistics*

- Equivalency: Master's in Probability — Statistics.

### **University of Yaoundé I**

**Yaoundé, Cameroon**

*Master's Degree in Mathematics*

- Equivalency: Specialized Higher Studies Diploma (DESS) in Mathematics.

## **Licenses & Certifications**

---

- AWS Certified Cloud Practitioner (CLF-C01)  
*Certification Prep Path (Pluralsight), Montréal*
- AWS Certified Developer — Associate (DVA-C02)  
*Certification Prep Path (Pluralsight), Montréal*

## **Conferences & Seminars**

---

### **PolyCongrès 2026, Polytechnique Montréal**

**Montréal, Canada**

*Oral Presentation*

*Mar–Apr 2026*

"Energy-Efficient and Near Real-Time Routing for Disaster Monitoring in Wireless Sensor Networks — Graph-Attentive Policy Fusion (GAPF)."

### **20th Conf. African Assoc. of Insect Scientists**

**Yaoundé, Cameroon**

*Oral Presentation*

*Oct 2013*

"Biogeography of communities of insects using clustering methods."

### **29th Annual Conf. Int. Society for Clinical Biostatistics**

**Copenhagen, Denmark**

*Poster*

*Aug 2008*

"HIV incidence estimation from repeated cross-sectional surveys."

### **ANMSA (African Network of Mathematical Statistics and its Applications)**

**Franceville, Gabon**

*Oral Presentation*

*Jan 2008*

"HIV epidemiology in Cameroon: Estimating incidence from prevalence data."

## **Volunteering**

---

- Volunteer at the IEEE branch of Polytechnique Montréal, Montréal, 2018.
- Volunteer at the ConFoo conference, Montréal, 2018.